

Multiple Choice

1. **Answer:** D

Options: A) electromagnetic radiation.; B) thermal conduction.; C) gravitational pressure.; D) radioactivity.; E) nuclear fusion.

Note: Radioactivity is a principal source of the Earth's internal energy because the decay of radioactive isotopes in the Earth's mantle and crust generates heat, contributing significantly to geothermal energy and driving geological processes.

2. **Answer:** B

Options: A) Gluon; B) Electron; C) Quark; D) Proton; E) Photon

Note: The negative muon (μ^-) is a leptonic particle with a charge of -1 and a similar spin and lepton number.

2. **Answer:** A

Options: A) 7.21×10^{25} nuclei; B) 9.09×10^{-26} nuclei; C) 4.48×10^{-26} nuclei; D) 26.98×10^{-27} nuclei; E) nuclei

Note: The correct answer is derived from calculating the number of atoms in 1 kg of aluminum by using its molar mass (approximately 27 g/mol) and Avogadro's number (approximately 6.022×10^{23} nuclei/mol), resulting in approximately 7.21×10^{25} nuclei.

3. **Answer:** A

Options: A) 0.75; B) 0.40; C) 0.55; D) 0.65; E) 0.50

Note: The coefficient of static friction is calculated using the formula $f = \mu_s \cdot N$, where f is the force of friction (343 N) and N is the normal force (equal to the weight of the mass, which is $50 \text{ kg} \times 9.8 \text{ m/s}^2 = 490 \text{ N}$), leading to $\mu_s = \frac{f}{N} = \frac{343 \text{ N}}{490 \text{ N}} = 0.75$.

4. **Answer:** A

Options: A) From accretion within the solar nebula at the same time the planets formed; B) From fragments of planets ejected by impacts; C) From dust grains that escape from passing comets; D) From the remnants of a planet's atmosphere; E) From the gravitational pull of nearby planets

Note: Planetary rings are formed from the leftover material in the solar nebula that did not coalesce into planets, with this process occurring concurrently with planet formation, leading to the accumulation of dust and ice particles that create the rings observed today.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the centripetal force required for a car to turn

depends on its speed and the radius of the turn, and in many scenarios, this force can be comparable to or greater than the car's weight, especially at high speeds or tight turns.

7. **Answer:** False

Options: A) True; B) False

Note: A minimum voltage of approximately 70,000 to 100,000 volts is typically required to produce K radiation from a tungsten target in an x-ray tube, making 65,000 volts insufficient.

8. **Answer:** True

Options: A) True; B) False

Note: The angular acceleration is calculated by converting the final angular velocity from revolutions per minute to radians per second and then dividing by the time taken to reach that velocity, which yields an angular acceleration of approximately 1.74 sec^{-2} .

8. **Answer:** True

Options: A) True; B) False

Note: This statement is true because the energy radiated per second per unit area by a blackbody is proportional to the fourth power of its absolute temperature, as described by the Stefan-Boltzmann Law, so increasing the temperature by a factor of 3 results in an increase in energy by a factor of $3^4 = 81$.

9. **Answer:** True

Options: A) True; B) False

Note: The statement is true because, according to Ampère's circuital law, a current-carrying wire generates a magnetic field that forms concentric circular loops around the wire, with the direction of the field lines being perpendicular to the axis of the wire.

Long Form Response

11. **Answer:** The relationship between voltage (V) and energy dissipation in a resistor is described by Joule's law, which states that the power (P) dissipated as heat in a resistor is proportional to the square of the voltage across it, given by the equation $P = V^2/R$, where R is the resistance. If the voltage across the resistor is doubled, the new power dissipation becomes $P' = (2 V)^2/R = 4 V^2/R$, indicating that the rate of energy dissipation quadruples. Thus, if the original power dissipation was 2 W, it would increase to 8 W when the voltage is doubled. This analysis highlights the quadratic relationship between voltage and energy dissipation, emphasizing the significant impact of voltage changes on power outputs in resistive components.

Note: The answer correctly identifies that power dissipation in a resistor is proportional to the square of the voltage across it, demonstrating that doubling the voltage leads to a quadrupling of the power dissipation due to the relationship expressed in Joule's law, $P = V^2/R$.

12. **Answer:** The time period of each oscillation of an ultrasonic transducer operating at 6.7 MHz can be calculated using the formula $T = 1/f$, where T is the time period and f is the frequency. Substituting the given frequency, $T = 1/(6.7 \times 10^6 \text{ Hz})$ results in approximately $1.49 \times 10^{-7} \text{ seconds}$, or $0.149 \text{ microseconds}$. To derive the angular frequency in rad/s , we use $2\pi f$. Therefore, substituting the frequency, $= 2(6.7 \times 10^6)$ gives an angular frequency of approximately 42,000 radians per second. This calculation highlights the inverse relationship between frequency and time period, while also illustrating the conversion from frequency to angular frequency in oscillatory systems.
- Note:* correct because it accurately applies the relationships between frequency, time period, and angular frequency, demonstrating the calculations necessary to derive both the time period of the ultrasonic transducer's oscillation and its corresponding angular frequency in radians per second.

End of Answer Key